

Preliminary,

The Zorn's Lemma gives several facts

- 1). Given a field  $F$ , there exists the Algebraic Closure  $\overline{F}$ .
- 2). Let  $F$  be a field and  $E/F$  is an Algebraic extension.  
Given an Embedding  $\varphi: F \rightarrow L$  where  $L$  is algebraically closed field,  
there exists an Extension  $\tilde{\varphi}: E \rightarrow L$  of  $\varphi$ .  
Moreover, if  $E$  is algebraically closed, such  $\tilde{\varphi}$  is Isomorphism.

In general, field Homomorphisms don't need to behave 'good' properties.

For instance, Let  $K = \mathbb{Q}(\sqrt[3]{2})$ . The minimal polynomial  $x^3 - 2$  over  $\mathbb{Q}$  of  $\sqrt[3]{2}$  has three distinct roots,  $\sqrt[3]{2}$ ,  $\omega \cdot \sqrt[3]{2}$ ,  $\omega^2 \cdot \sqrt[3]{2}$  where  $\omega$  is primitive 3<sup>th</sup> root of unity. Therefore,  $\# \text{Emb}_{\mathbb{Q}}(\mathbb{Q}(\sqrt[3]{2}), \mathbb{C}) = [\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$ .

But, since  $\omega^i \sqrt[3]{2} \notin \mathbb{Q}(\sqrt[3]{2})$   $i=1,2$ , so

$$\text{Aut}(\mathbb{Q}(\sqrt[3]{2})) = \{\text{id}\},$$

